MILITARY STANDARD

MIL-STD-1320-202
(NAVY)

29 FEBRUARY 1980

TRUCKLOADING OF HAZARDOUS MATERIALS

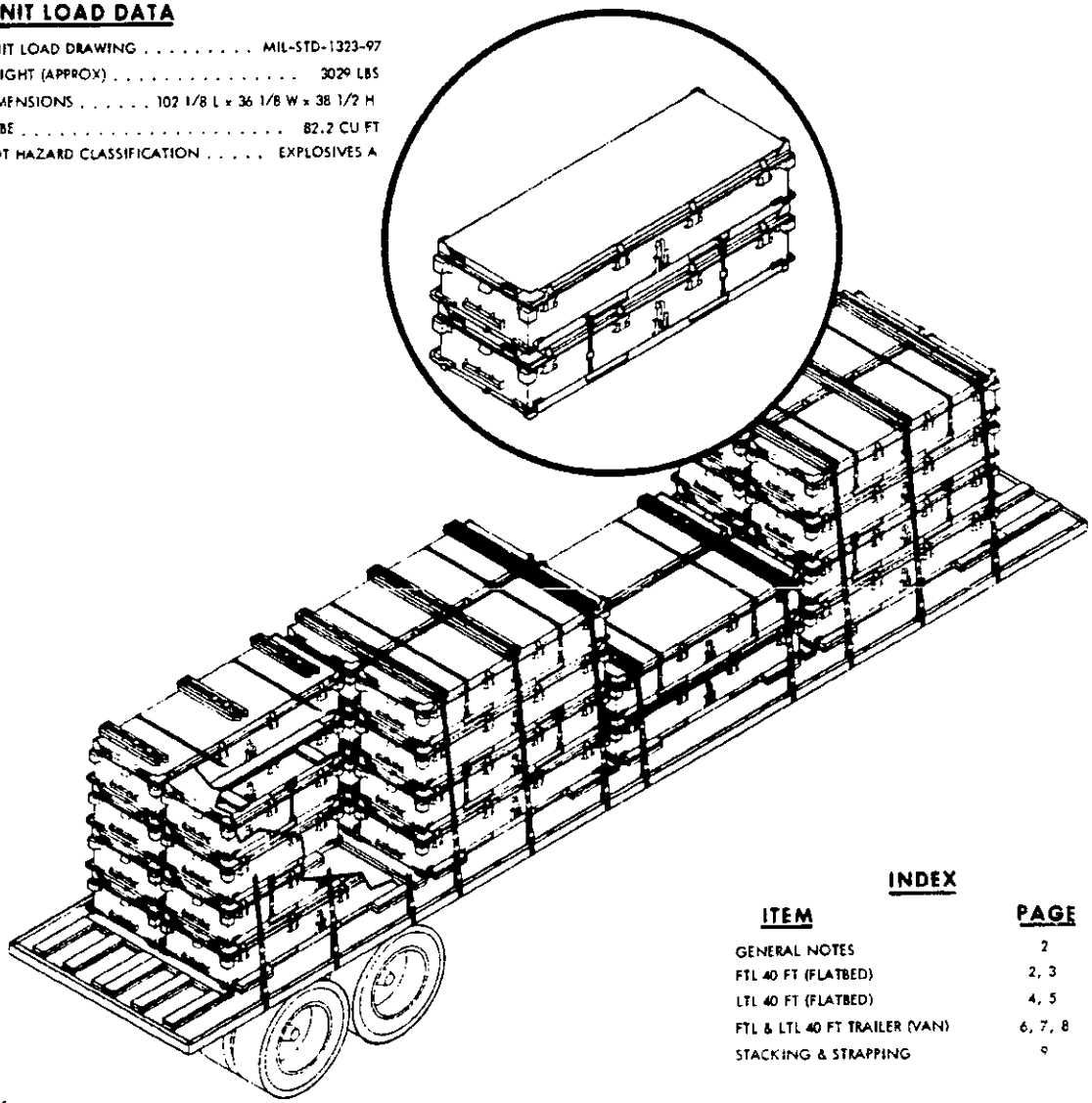
DISPENSER AND BOMB AIRCRAFT CBU-MK 20 & MODS (ROCKEYE II)

IN CONTAINER CNU-319/E

UNIT LOAD UNDERWAY REPLENISHMENT

UNIT LOAD DATA

UNIT LOAD DRAWING MIL-STD-1323-97
WEIGHT (APPROX) . . . . . 3029 LBS
DIMENSIONS . . . . . 102 1/8 L x 36 1/8 W x 38 1/2 H
CUBE . . . . . 82.2 CU FT
DOT HAZARD CLASSIFICATION . . . . . EXPLOSIVES A



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NOTES:
1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, CONTAINER LOADING AND CARLOADING MILITARY STANDARDS REFER TO INDEX TO STANDARDS, MIL-HDBK-236 (NAVY).

FSC 8140

AUTHORIZED AND RELEASED FOR HIGHWAY SHIPMENTS ONLY

SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE

SIGNATURE AIRSYSCOM BY DIRECTION DATE

ORIGINATOR

DATE

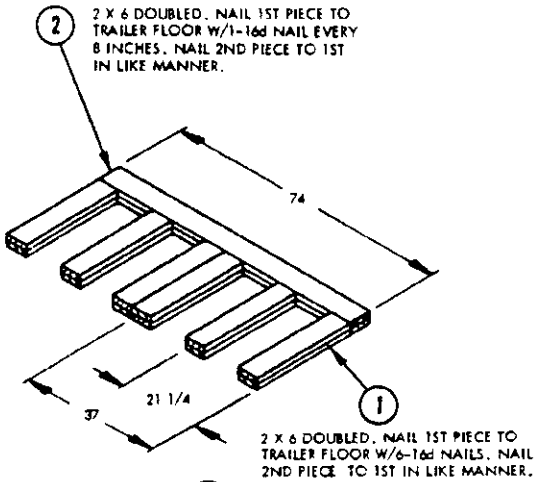
NAVAL WEAPONS HANDLING CENTER
WPNSTA EARLE, NEW JERSEY

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GENERAL NOTES

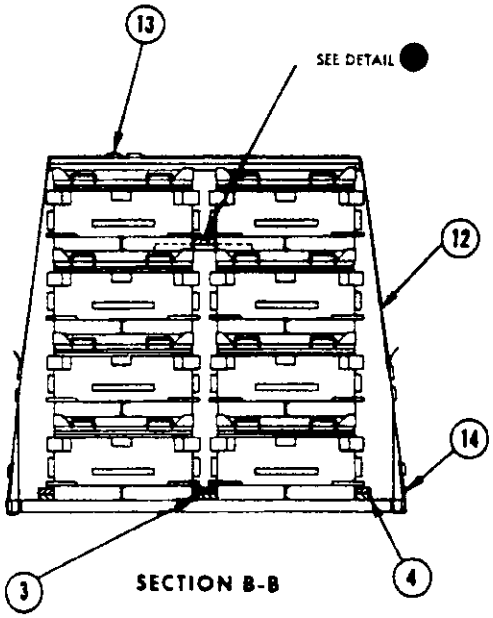
1. THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR TRUCKLOADING UNIT LOADS FOR UNDERWAY REPLENISHMENT OF THE DISPENSER AND BOMB AIRCRAFT, CBU-MK 20 AND MODS (ROCKEYE II) IN CONTAINER CNU-319/E WHEN THE CONTAINERS ARE UNITIZED IN ACCORDANCE WITH MIL-STD-1323-97 (NAVY).
2. THE PROCEDURES AND PRACTICES DESCRIBED HEREIN ARE INTENDED FOR 40 FT FLATBED TRAILERS OR 40 FT VAN TRAILERS. THE TRAILER AXLES SHALL BE LOCATED IN THE WESTERN POSITION (AT THE EXTREEM REAR OF THE TRAILER). DO NOT USE TRAILERS WITH NON-NAILABLE FLOORS.
3. WHEN REQUIRED TO SHIP UNIT LOADS TWO HIGH, THEY SHALL BE STACKED AND STRAPPED TOGETHER AS SPECIFIED ON PAGE 9.
4. EACH STACK OF CONTAINERS SHALL BE SECURED TOGETHER WITH TWO 1 1/4 X .035 CROSS STRAPS. THE CROSS STRAP SHALL PASS OVER THE COVERS OF THE CONTAINERS IN THE TOP LAYER AND RETURN PASSING THROUGH THE FORK POCKETS OF THESE CONTAINERS, BINDING THE TOP LAYER TOGETHER.
5. CHAINS AND STEEL STRAPS MAY BE USED INTERCHANGEABLY FOR TIE-DOWNS ON A 1 TO 1 BASIS.
6. STEEL STRAP USED AS TIE-DOWNS SHALL BE 2 X .050 AND SHALL CONFORM TO AND BE APPLIED AS SPECIFIED IN THE BASIC TRUCKLOADING DOCUMENT MIL-STD-1320 (NAVY) AND AS SHOWN ON PAGE 3 OF THIS DOCUMENT.
7. CHAINS, FITTINGS AND LOADBINDERS SHALL MEET THE REQUIREMENTS OF THE BASIC TRUCKLOADING DOCUMENT MIL-STD-1320 (NAVY) AND APPLIED AS SHOWN ON PAGE 5 OF THIS DOCUMENT.
8. PRIOR TO LOADING THE TRAILER AND DURING PRELOADING INSPECTION REQUIRED BY NAVWEPS OP 2165 AND REPORTED ON DD FORM 626, THE CHAINS, FITTINGS AND LOAD BINDERS SHALL BE INSPECTED FOR STRETCH, GOUGING, BENT LINKS, WEAR AND ANY OTHER NOTICEABLE DEFECTS. THE INSPECTOR SHALL CONFIRM THAT THE CHAINS AND LOADBINDERS HAVE BEEN INSPECTED AND SHALL SO NOTE ON ITEM NO. 22 OF DD FORM 626. ANY DEFICIENCY SHALL BE CAUSE FOR REJECTION OF A CHAIN OR LOADBINDER.
9. THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILTY OF THE CARRIER. THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS.
10. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED, ATTACH SHIPPING DOCUMENTS TO DUNNAGE IN AN ACCESSIBLE AREA AND ATTACH PLACARD "EXPLOSIVES A" TO BOTH SIDES, FRONT AND BACK OF TRAILER.
11. UNLESS OTHERWISE SPECIFIED, ALL MATERIAL SHALL BE AS SPECIFIED IN THE BASIC TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY).
12. FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1370 (NAVY).

FTL 40 FT TRAILER (FLATBED)



DETAIL A

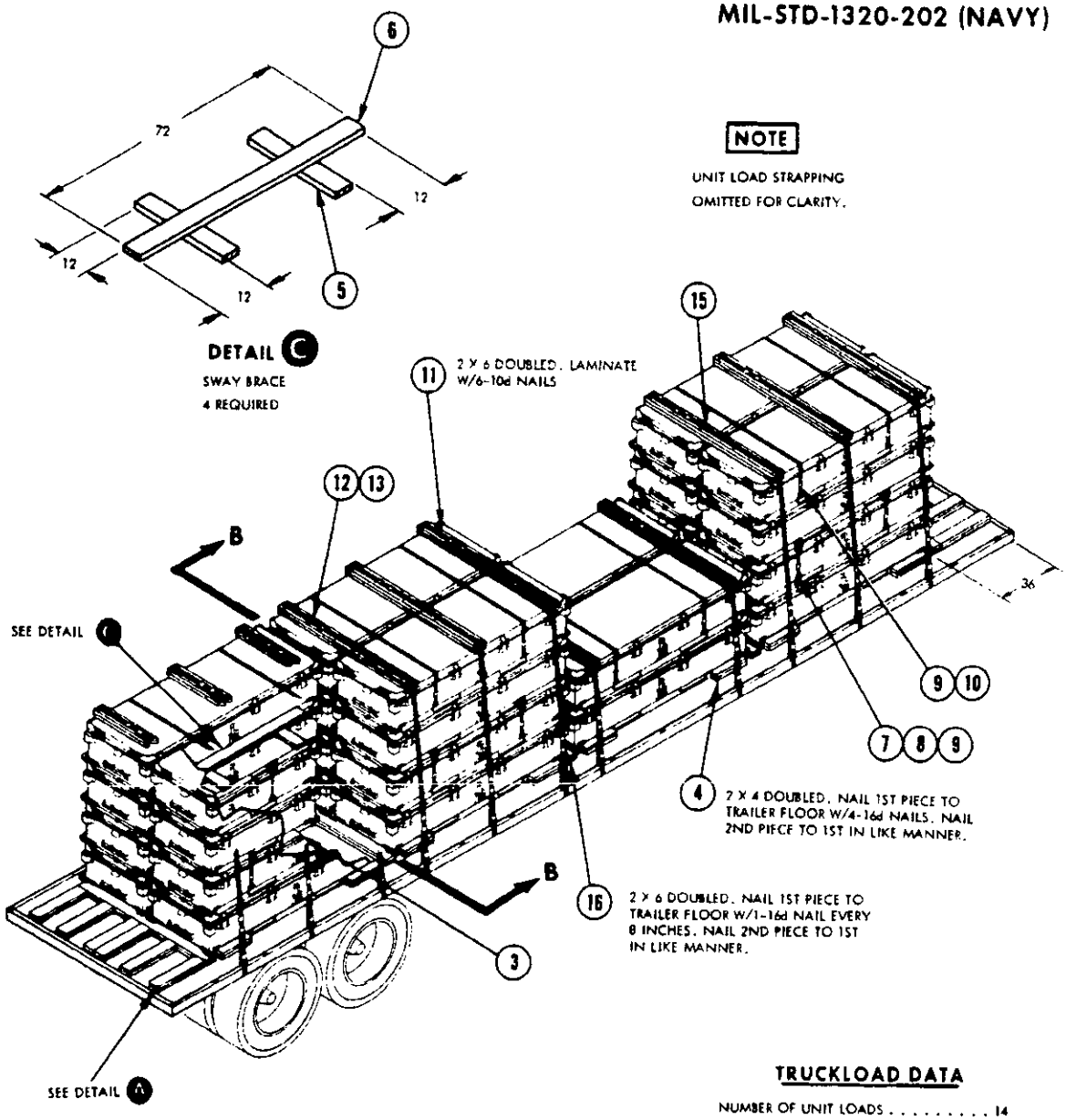
FRONT AND REAR BLOCKING (ASSEMBLE IN PLACE)



SECTION B-B

2 X 6 DOUBLED. NAIL 1ST PIECE TO TRAILER FLOOR W/1-16d NAIL EVERY 8 INCHES. NAIL 2ND PIECE TO 1ST IN LIKE MANNER.

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NOTE

UNIT LOAD STRAPPING OMITTED FOR CLARITY.

TRUCKLOAD DATA

NUMBER OF UNIT LOADS 14

TRUCKLOAD WEIGHT (APPROX) . . 42,406 LBS.

PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NAILS NUMBER	NAILS SIZE
16	INTERMEDIATE CROSSMEMBER	2 X 6 X 74	6	SEE FIELD NOTE		
15	STAPLE	FOR 2" STRAP	44	-	-	-
14	STAKE POCKET PAD (STRAPPING)	2 X .050 X 24	22	-	-	-
13	SEAL	FOR 2" STRAP	88	-	-	-
12	TIEDOWN STRAP	2 X .050 X 20 FT	11	-	-	-
11	PROTECTOR BOARD	2 X 6 X 72	22	-	-	-
10	CROSS STRAP	1 1/4 X .035 X 18 FT	8	-	-	-
9	SEAL	FOR 1 1/4 STRAP	20	-	-	-

LIST OF MATERIALS & NAILING DATA

PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NAILS NUMBER	NAILS SIZE
8	EDGE PROTECTOR STEEL STRAP	NSN 8135-00-291-0962	24	-	-	-
7	UNITIZING STRAP	1 1/4 X .035 X 9 FT 6	12	-	-	-
6	SPACER	2 X 6 X 72	4	5	3/ JOINT	16d CLND
5	SUPPORT	2 X 6 X 30	8	SEE 6	-	-
4	SLEEPER	2 X 4 X 18	32	SEE FIELD NOTE		
3	SPACER	2 X 6 X 72	8	SEE SECTION B-B		
2	END CROSSMEMBER	2 X 6 X 74	4	SEE DETAIL A		
1	BACKUP CLEAT	2 X 6 X 28	24	SEE DETAIL A		

LIST OF MATERIALS & NAILING DATA

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LTL 40 FT TRAILER (FLATBED)

1. WHEN REQUIRED TO SHIP 9, 10, 11, 12 OR 13 UNIT LOADS, SELECT THE CORRECT LOAD PATTERN FOR THE NUMBER OF UNIT LOADS TO BE SHIPPED FROM TABLE I AND POSITION THEM ON THE TRAILER ACCORDINGLY.

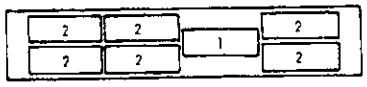
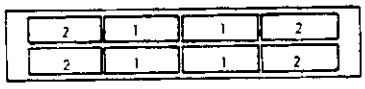
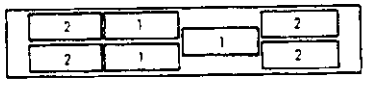
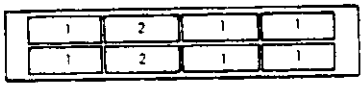
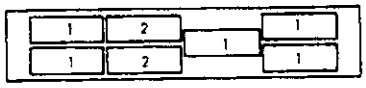
NOTE

WHEN REQUIRED TO SHIP 8 OR LESS UNIT LOADS, THEY MAY BE LOADED ON A FLATBED TRAILER, DUNNAGE AND TIE DOWN USING THE PRINCIPLES SHOWN ON PAGE 5 FOR THE 1 HIGH UNIT LOADS.

NOTE

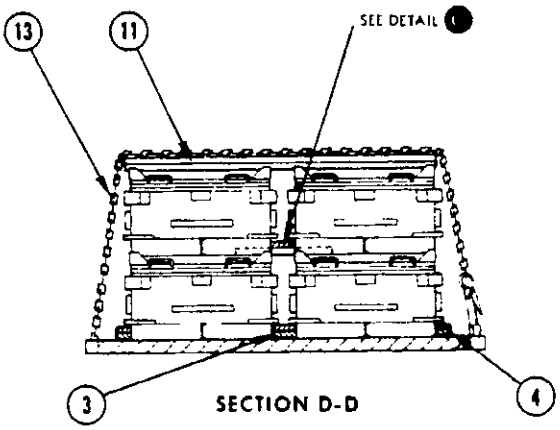
8 OR LESS UNIT LOADS MAY ALSO BE SHIPPED IN A VAN TRAILER. SEE PAGES 7 AND 8.

TABLE I

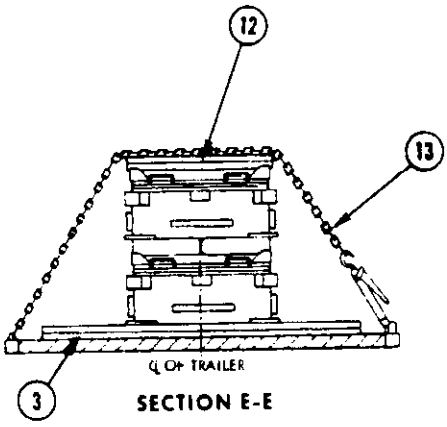
NO. OF UNIT LOADS	LOAD PATTERN *	FWD
13		
12		
11		
10		
9		

* LOAD PATTERN SHOWS THE FLOOR PLAN FOR THE LOADED TRAILER. NUMBER IN LOAD PATTERN BLOCKS INDICATES THE NUMBER OF LAYERS OF UNIT LOADS OF CONTAINERS.

2. THE LTL ON PAGE 5 IS FOR 11 UNIT LOADS. LTL SHIPMENTS ARE NOT LIMITED TO THIS QUANTITY. ACTUAL SHIPPING REQUIREMENTS WILL DETERMINE THE NUMBER OF UNIT LOADS TO BE SHIPPED AND THEY SHOULD BE DUNNAGED AND TIED DOWN IN ACCORDANCE WITH APPROPRIATE PARTS OF THIS STANDARD.



SECTION D-D



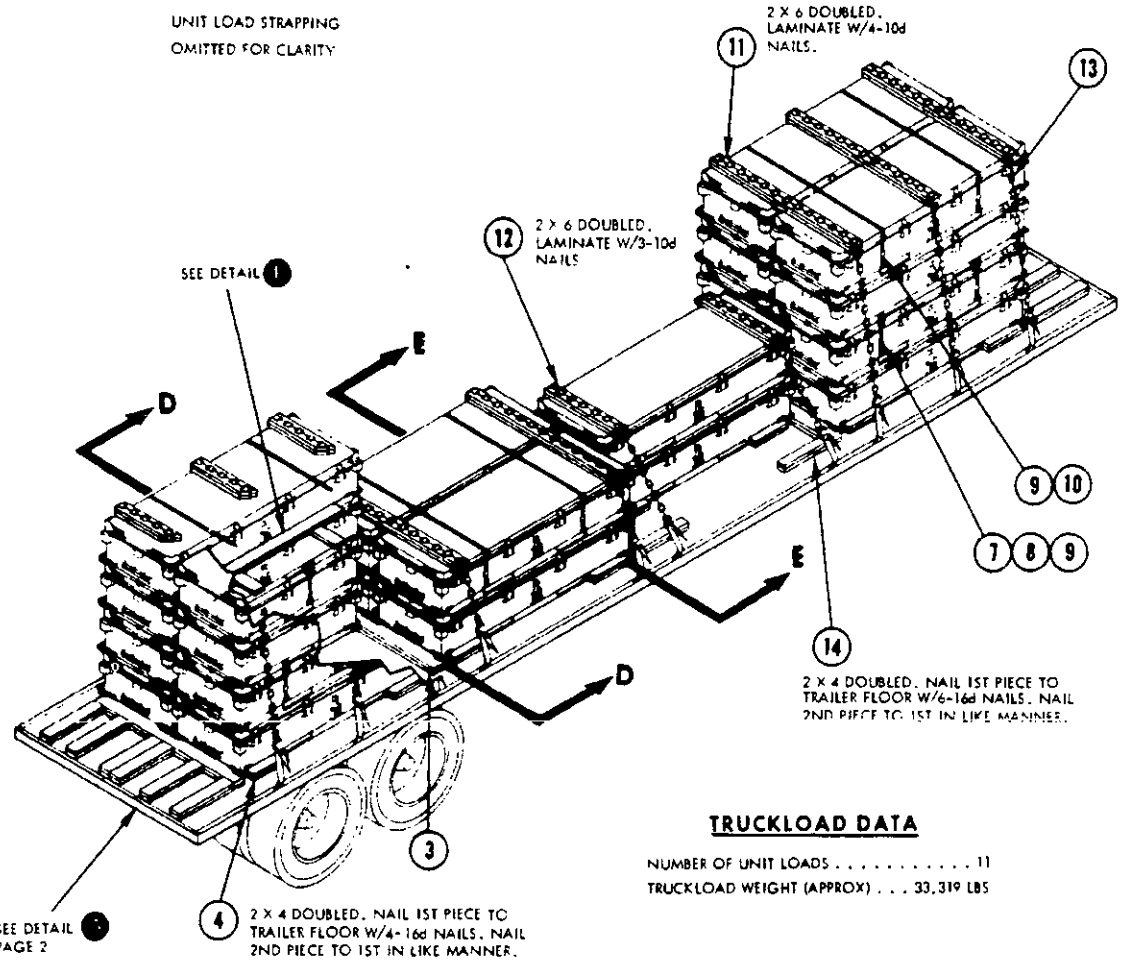
SECTION E-E

2 X 6 DOUBLED. NAIL 1ST PIECE TO TRAILER FLOOR W/1-16d NAIL EVERY 8 INCHES. NAIL 2ND PIECE TO 1ST IN LIKE MANNER.

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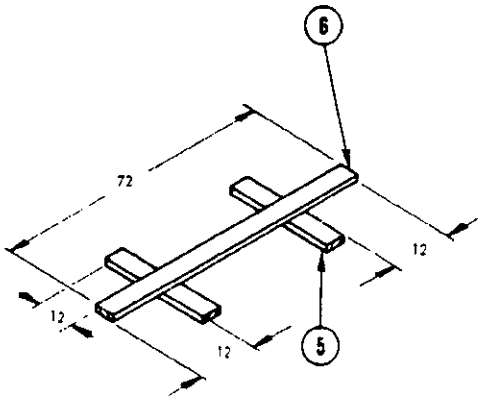
NOTE

UNIT LOAD STRAPPING
OMITTED FOR CLARITY



TRUCKLOAD DATA

NUMBER OF UNIT LOADS 11
TRUCKLOAD WEIGHT (APPROX) . . . 33,319 LBS



DETAIL F
SWAY BRACE
4 REQUIRED

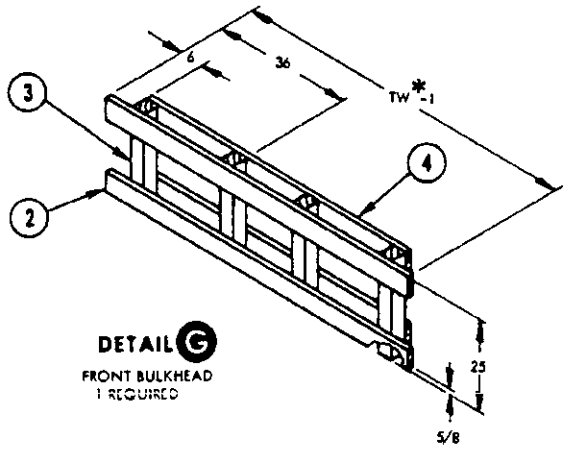
PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NUMBER	SIZE
					NAILS	
14	BACKUP CLEAT	2 X 4 X 24	8	SEE FIELD NOTE		
13	CHAIN & LOAD BINDER	5/16 OR 3/8	10	-	-	-
12	PROTECTOR BOARD	2 X 6 X 34	4	-	-	-
11	PROTECTOR BOARD	2 X 6 X 72	16	-	-	-
10	CROSS STRAP	1 1/4 X .035 X 18 FT	6	-	-	-
9	SEAL	FOR 1 1/4 STRAP	14	-	-	-
8	EDGE PROTECTOR, STEEL STRAP	NSN: 8135-00-291-0962	16	-	-	-
7	UNITIZING STRAP	1 1/4 X .035 X 9 FT 6	8	-	-	-
6	SPACER	2 X 6 X 72	3	5	3/JOINT	16d CLND
5	SUPPORT	2 X 6 X 30	6	SEE 6	-	-
4	SLEEPER	2 X 4 X 18	32	SEE FIELD NOTE		
3	SPACER	2 X 6 X 24	6	SEE SECTION D-D		
2	END CROSSMEMBER	2 X 6 X 24	4	SEE DETAIL A		
1	BACKUP CLEAT	2 X 6 X 28	24	SEE DETAIL A		

LIST OF MATERIALS & NAILING DATA

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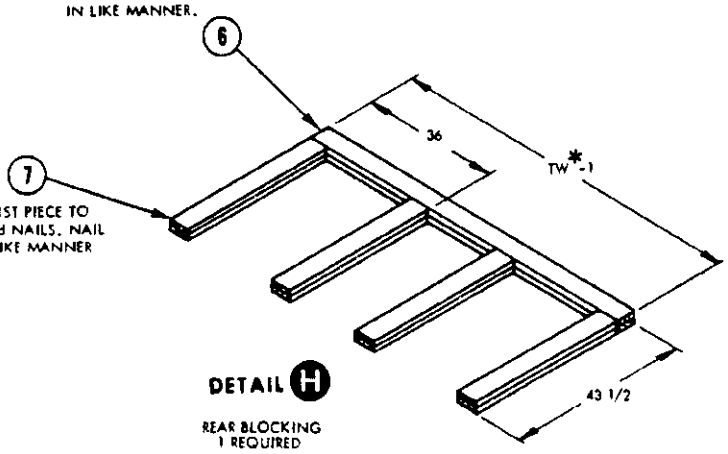
FTL AND LTL 40 FT TRAILER (VAN)

1. A UNIT LOAD OF CLUSTER BOMBS IN CNU-319/E CONTAINERS IS LOADED INTO A VAN USING A 6000 LB CAPACITY FORKLIFT TRUCK. HANDLING THE UNIT LOAD FROM THE END.
2. A FTL VAN CONSISTS OF 8 UNIT LOADS AND IS LOADED, BLOCKED AND BRACED AS SHOWN ON PAGE 7. WHEN REQUIRED TO SHIP LESS THAN 8 UNIT LOADS IN A VAN, LOAD, BLOCK AND BRACE FOLLOWING THE PRINCIPLES OF THE FTL VAN ON PAGE 7 AND THE LTL VAN ON PAGE 8.



DETAIL G
FRONT BULKHEAD
1 REQUIRED

2 X 6 DOUBLED. NAIL 1ST PIECE TO TRAILER FLOOR W/1-16d NAIL EVERY 8 INCHES. NAIL 2ND PIECE TO 1ST IN LIKE MANNER.

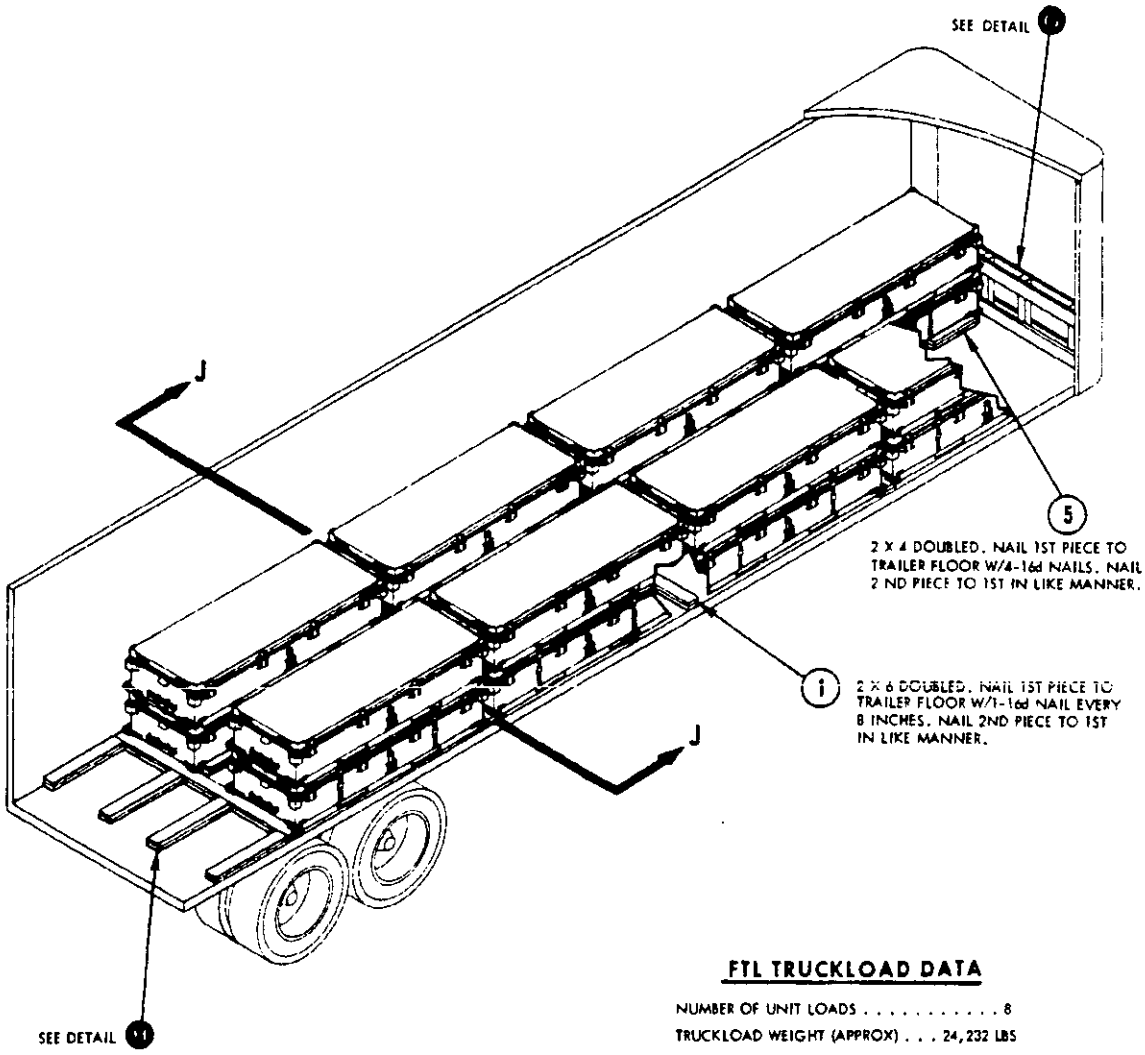


2 X 6 DOUBLED. NAIL 1ST PIECE TO TRAILER FLOOR W/6-16d NAILS. NAIL 2ND PIECE TO 1ST IN LIKE MANNER

DETAIL H
REAR BLOCKING
1 REQUIRED
ASSEMBLE IN PLACE

* TW = TRAILER WIDTH

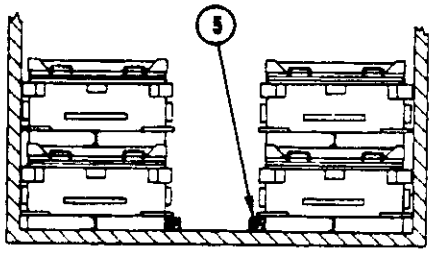
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FTL TRUCKLOAD DATA

NUMBER OF UNIT LOADS 8

TRUCKLOAD WEIGHT (APPROX) . . . 24,232 LBS



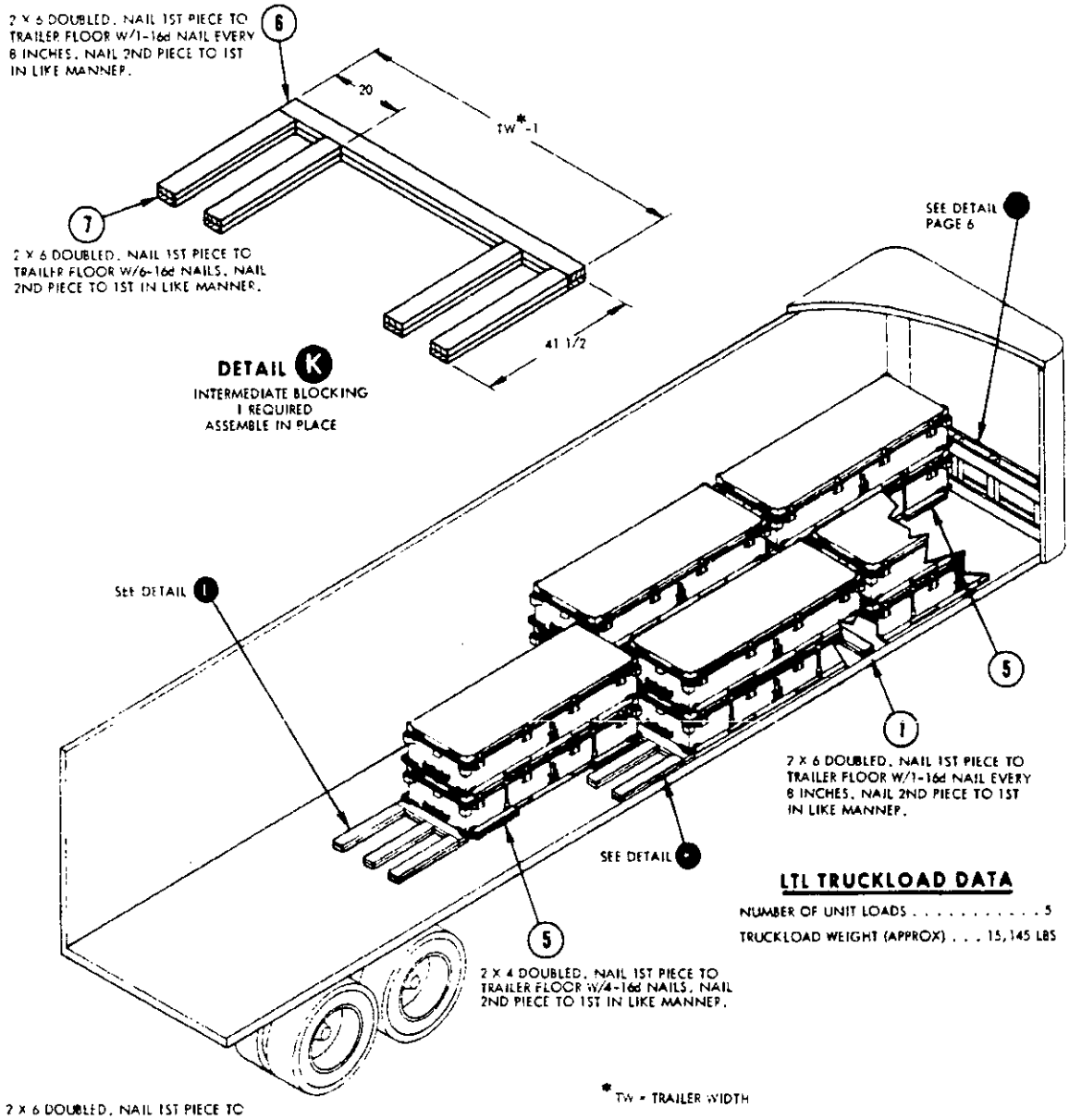
SECTION J-J

* TW = TRAILER WIDTH

PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NUMBER NAILS	SIZE
7	BACKUP CLEAT	2 X 6 X 38	8	SEE DETAIL H		
6	CROSSMEMBER	2 X 6 X TW*-1	2	SEE DETAIL H		
5	SLEEPER	2 X 4 X 24	32	SEE FIELD NOTE		
4	FWD CROSSMEMBER	2 X 6 X TO SUIT	2	3	3/JOINT	12d
3	VERTICAL	4 X 4 X 25	4	SEE 2 & 4	-	-
2	AFT CROSSMEMBER	2 X 6 X TW*-1	2	3	3/JOINT	12d
1	SPACER CROSSMEMBER	2 X 6 X TW*-1	6	SEE FIELD NOTE		

LIST OF MATERIALS & NAILING DATA

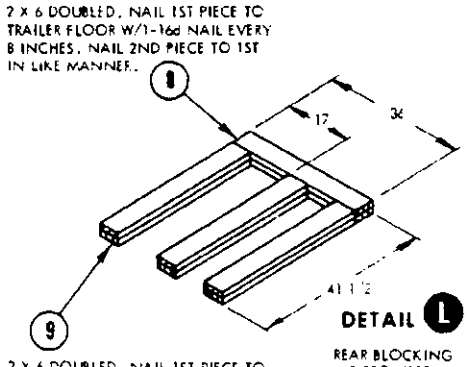
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LTL TRUCKLOAD DATA

NUMBER OF UNIT LOADS 5

TRUCKLOAD WEIGHT (APPROX) . . . 15,145 LBS



DETAIL L

REAR BLOCKING
1 REQUIRED
ASSEMBLE IN PLACE

*TW = TRAILER WIDTH

PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NUMBER	SIZE
					NAILS	
9	BACKUP CLEAT	2 X 6 X 36	6	SEE DETAIL L		
8	REAR CROSSMEMBER	2 X 6 X 36	2	SEE DETAIL L		
7	BACKUP CLEAT	2 X 6 X 36	8	SEE DETAIL K		
6	INTERMEDIATE CROSSMEMBER	2 X 6 X TW*-1	2	SEE DETAIL K		
5	SLEEPER	2 X 4 X 24	24	SEE FIELD NOTE		
4	FWD CROSSMEMBER	2 X 6 X TO SUIT	2	3	3"JOINT	12d
3	VERTICAL	4 X 4 X 25	4	SEE 2 & 4	-	-
2	AFT CROSSMEMBER	2 X 6 X TW*-1	2	3	3"JOINT	12d
1	SPACER CROSSMEMBER	2 X 4 X TW*-1	2	SEE FIELD NOTE		

LIST OF MATERIALS & NAILING DATA

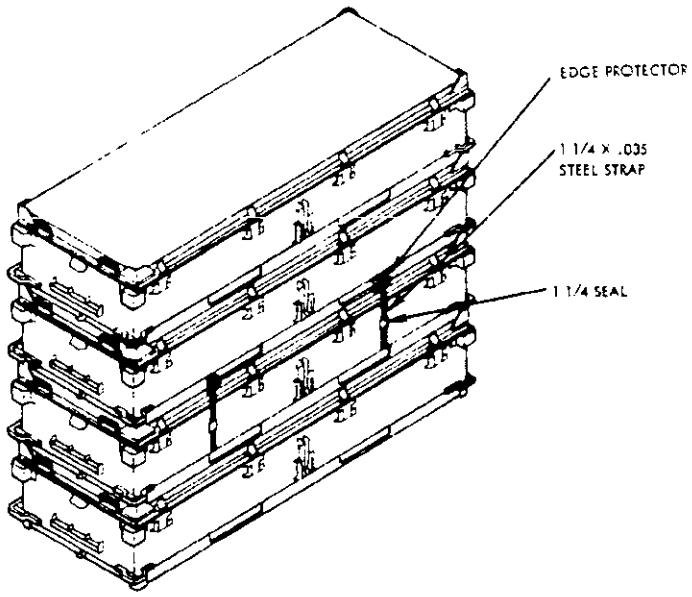
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STRAPPING AND STACKING PROCEDURE

1. STACK THE UNIT LOADS TOGETHER USING CARE TO ALIGN THE INTERLOCKING FEATURE OF THE CONTAINER.
2. THREAD THE STRAPPING THROUGH THE FORK POCKETS OF THE TOP CONTAINER OF THE BOTTOM UNIT LOAD AND THE BOTTOM CONTAINER OF THE TOP UNIT LOAD. TENSION AND SECURE WITH ONE DOUBLE NOTCHED SEAL OR TWO DOUBLE CRIMPED SEALS.
3. STACKED AND STRAPPED UNIT LOADS OF CNU-319E CONTAINERS ARE HANDLED AND POSITIONED ON FLATBED TRAILERS USING FORK-LIFT TRUCKS OF SUITABLE CAPACITY. IF ADEQUATE CAPACITY EQUIPMENT IS NOT AVAILABLE, THE UNIT LOADS SHOULD BE POSITIONED ON THE TRAILER ONE AT A TIME AND THE STACKED UNIT LOADS STRAPPED TOGETHER IN POSITION.

WARNING

WHEN LOADING/UNLOADING TRAILERS WITH UNITIZED CONTAINERS, EXTRA CAUTION SHOULD BE TAKEN TO PREVENT TOPPLING. STACKED UNIT LOADS SHALL BE BROKEN DOWN TO UNIT LOADS AFTER UNLOADING THE VEHICLE.



NOTE

UNIT LOAD STRAPPING OMITTED FOR CLARITY.

REVIEW ACTIVITY:
NAVY - OS, AS

PREPARING ACTIVITY:
NAVY - OS
(PROJECT NO. 8140-N464)

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